1 Latent Regression Model

We consider C classes (or datasets), indexed by $c = 1, \dots, C$. For each class c, we observe a data matrix $X^{(c)} \in \mathbb{R}^{N_c \times M}$, where each row $\boldsymbol{x}_i^{(c)} \in \mathbb{R}^M$ is associated with a D-dimensional latent variable $\boldsymbol{z}_i^{(c)} \in \mathbb{R}^D$.

Thus, each class is associated with a collection of latent variables

$$\boldsymbol{z}^{(c)} = \{\boldsymbol{z}_i^{(c)}\}_{i=1}^{N_c}, \qquad \boldsymbol{z}_i^{(c)} \in \mathbb{R}^D.$$

For each class c, the latent variables are assumed independent and identically distributed according to

$$\boldsymbol{z}_i^{(c)} \sim \mathcal{N}\big(\boldsymbol{\mu}^{(c)}, \tau_c^{-1} I_D\big), \qquad c = 2, \dots, C,$$

where $\boldsymbol{\mu}^{(c)} \in \mathbb{R}^D$ is the class-specific latent mean and $\tau_c > 0$ is a precision parameter controlling the latent dispersion.

To remove redundancy, we fix the latent distribution of the first class as a reference coordinate system. Specifically, we impose

$$\boldsymbol{\mu}^{(1)} = \mathbf{0}, \qquad \tau_1 = 1,$$

so that

$$\boldsymbol{z}_i^{(1)} \sim \mathcal{N}(\mathbf{0}, I_D).$$

This anchoring removes global translation and scaling symmetries without restricting the expressive capacity of the model. The latent geometry of each class is therefore interpreted relative to the reference class $c = 1$.

Let $\phi : \mathbb{R}^D \to \mathbb{R}^K$ denote a fixed nonlinear basis mapping. The design matrix of class c is defined as $Z^{(c)} \in \mathbb{R}^{N_c \times K}$ with entries

$$Z_{ik}^{(c)} = \phi_k(\boldsymbol{z}_i^{(c)}).$$

We collect the decoder weights in a shared matrix

$$W = [\boldsymbol{w}_1 \ \dots \ \boldsymbol{w}_M] \in \mathbb{R}^{K \times M},$$

where each column $\boldsymbol{w}_j \in \mathbb{R}^K$ corresponds to the weights for the j-th observed dimension across all classes.

Conditioned on $\boldsymbol{z}^{(c)}$ and W, observations are generated as

$$p(X^{(c)} \mid \boldsymbol{z}^{(c)}, W) = \prod_{i=1}^{N_c} \prod_{j=1}^{M} \mathcal{N}\left(x_{ij}^{(c)} \;\middle|\; \phi(\boldsymbol{z}_i^{(c)})^\top \boldsymbol{w}_j, \beta^{-1}\right),$$

or equivalently,

$$\boldsymbol{x}_j^{(c)} \sim \mathcal{N}\big(Z^{(c)}\boldsymbol{w}_j, \beta^{-1} I_{N_c}\big)\,,$$

where $\beta > 0$ is the observation precision and $\boldsymbol{x}_j^{(c)}$ denotes the j-th column of $X^{(c)}$.

To regularize the decoder, we assume noninformative priors for the latent parameters $\{\boldsymbol{\mu}^{(c)}, \tau_c\}_{c=2}^{C}$, and place independent Gaussian priors on the columns of W:

$$p(W) = \prod_{j=1}^{M} \mathcal{N}\left(\boldsymbol{w}_j \;\middle|\; \mathbf{0}, \alpha_j^{-1} I_K\right).$$

The full joint distribution factorizes as

$$p(W) \prod_{c=1}^{C} p(X^{(c)} \mid \boldsymbol{z}^{(c)}, W) p(\boldsymbol{z}^{(c)} \mid \boldsymbol{\mu}^{(c)}, \tau_c) p(\boldsymbol{\mu}^{(c)}, \tau_c).$$

This defines a shared nonlinear generative model in which all classes are coupled through the common decoder W, while retaining class-specific latent geometry via $(\boldsymbol{\mu}^{(c)}, \tau_c)$.

1.1 Variational Inference for the Single-Class Case

To simplify the exposition, we now consider the single-class case. Since only one class is present, we remove the superscript (c) from the notation for clarity.

Furthermore, we restrict the latent variable to be scalar,

$$\boldsymbol{z} = \{z_i\}_{i=1}^{N} \qquad z_i \in \mathbb{R},$$

so that

$$p(\boldsymbol{z} \mid \mu, \tau) = \prod_{i=1}^{N} \mathcal{N}(z_i \mid \mu, \tau^{-1}).$$

We treat μ and τ as unknown parameters to be inferred, using the standard joint Jeffreys prior for the mean and precision of a Gaussian model,

$$p(\mu, \tau) \propto \tau^{-1}.$$

The nonlinear basis mapping reduces to

$$\phi : \mathbb{R} \to \mathbb{R}^K,$$

and the design matrix $Z \in \mathbb{R}^{N \times K}$ is defined by

$$Z_{ik} = \phi_k(z_i).$$

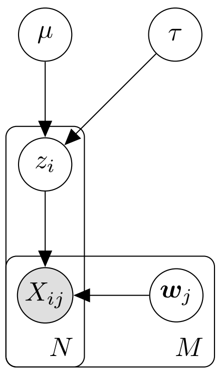

Figure 1: One class scalar latent regression model

We see the Bayesian network of this model in Figure 1. Exact posterior inference in this model is intractable due to the nonlinear dependence of the likelihood on $\boldsymbol{z}$. We therefore proceed by introducing a variational approximation.

In order to formulate a variational treatment of this model, we consider a variational distribution which factorizes between the latent variables and the parameters so that

$$q(\boldsymbol{z}, W, \mu, \tau) = q(\boldsymbol{z})q(W, \mu, \tau).$$

First, let us consider the derivation of the update equation for the factor $q(W, \mu, \tau)$. The log of the optimized factor is given by

$$\ln q^*(W, \mu, \tau) = \mathbb{E}_{q(\boldsymbol{z})}\left[\ln p(W, \mu, \tau \mid \boldsymbol{z}, X)\right] + \text{const}.$$

As can be seen from Figure 1, the d-separation criterion implies that

$$\mathbb{E}_{q(\boldsymbol{z})}\left[\ln p(W, \mu, \tau \mid \boldsymbol{z}, X)\right] = \mathbb{E}_{q(\boldsymbol{z})}\left[\ln p(\mu, \tau \mid \boldsymbol{z})\right] + \sum_j \mathbb{E}_{q(\boldsymbol{z})}\left[\ln p(\boldsymbol{w}_j \mid \boldsymbol{z}, X)\right].$$

Therefore, we have the following induced factorization for $q(W, \mu, \tau)$:

$$q(W, \mu, \tau) = q(\mu, \tau) \prod_{j=1}^{M} q(\boldsymbol{w}_j).$$

We obtain $q(\boldsymbol{w}_j)$ using the fact that each $\boldsymbol{w}_j$ is a weight vector of a Bayesian linear regression model:

$$\begin{aligned} \ln p(\boldsymbol{w}_j \mid \boldsymbol{z}, X) &= -\frac{\beta}{2}\left\|Z\boldsymbol{w}_j - \boldsymbol{x}_j\right\|^2 - \frac{\alpha_j}{2}\boldsymbol{w}_j^\top \boldsymbol{w}_j + \text{const} \\ &= -\frac{1}{2}\boldsymbol{w}_j^\top(\beta Z^\top Z + \alpha_j I)\boldsymbol{w}_j + \beta \boldsymbol{x}_j^\top Z \boldsymbol{w}_j + \text{const}. \end{aligned}$$

Taking the expectation with respect to $q(\boldsymbol{z})$, we obtain

$$\ln q^*(\boldsymbol{w}_j) = -\frac{1}{2}\boldsymbol{w}_j^\top(\beta\mathbb{E}_{q(\boldsymbol{z})}\left[Z^\top Z\right] + \alpha_j I)\boldsymbol{w}_j + \beta\boldsymbol{w}_j^\top\mathbb{E}_{q(\boldsymbol{z})}\left[Z\right]^\top \boldsymbol{x}_j + \text{const.}$$

The expression above is quadratic in $\boldsymbol{w}_j$. Completing the square yields

$$q^*(\boldsymbol{w}_j) = \mathcal{N}(\boldsymbol{w}_j \mid \boldsymbol{\mu}_j, \Sigma_j).$$

The corresponding moments are

$$\begin{aligned}\mathbb{E}_{q(\boldsymbol{w}_j)}\left[\boldsymbol{w}_j\right] &= \boldsymbol{\mu}_j,\\ \mathbb{E}_{q(\boldsymbol{w}_j)}\left[\boldsymbol{w}_j\boldsymbol{w}_j^\top\right] &= \boldsymbol{\mu}_j\boldsymbol{\mu}_j^\top + \Sigma_j,\end{aligned}$$

with

$$\begin{aligned}\Sigma_j^{-1} &= \beta\mathbb{E}_{q(\boldsymbol{z})}\left[Z^\top Z\right] + \alpha_j I,\\ \boldsymbol{\mu}_j &= \beta\Sigma_j\mathbb{E}_{q(\boldsymbol{z})}\left[Z\right]^\top \boldsymbol{x}_j.\end{aligned}$$

Collecting the columns, we obtain

$$\mathbb{E}_{q(W)}\left[W\right] = [\boldsymbol{\mu}_1 \ \ \ldots \ \ \boldsymbol{\mu}_M].$$

Furthermore,

$$\sum_j \mathbb{E}_{q(\boldsymbol{w}_j)}\left[\boldsymbol{w}_j\boldsymbol{w}_j^\top\right] = \mathbb{E}_{q(W)}\left[W\right]\mathbb{E}_{q(W)}\left[W\right]^\top + \sum_{j=1}^{M}\Sigma_j.$$

Note that if $\Sigma_j = \Sigma$ for all j, the mean simplifies to

$$\mathbb{E}_{q(W)}\left[W\right] = \beta\Sigma\mathbb{E}_{q(\boldsymbol{z})}\left[Z\right]^\top X.$$

We now consider the update equation for the factor $q(\mu, \tau)$:

$$\begin{aligned}\ln q^*(\mu, \tau) &= \mathbb{E}_{q(\boldsymbol{z})}\left[\ln p(\mu, \tau \mid \boldsymbol{z})\right] + \text{const}\\ &= -\frac{\tau}{2}\left(\sum_i \mathbb{E}_{q(z_i)}\left[z_i^2\right] - 2\mu\sum_i \mathbb{E}_{q(z_i)}\left[z_i\right] + N\mu^2\right)\\ &\quad + \left(\frac{N}{2} - 1\right)\ln\tau + \text{const.}\end{aligned}$$

We define

$$\begin{aligned}\overline{z} &= \frac{1}{N}\sum_i \mathbb{E}_{q(z_i)}\left[z_i\right],\\ \overline{zz} &= \frac{1}{N}\sum_i \mathbb{E}_{q(z_i)}\left[z_i^2\right].\end{aligned}$$

We observe that $q^*(\mu, \tau)$, for $N > 1$ follows a Normal–Inverse–Gamma distribution:

$$q^*(\mu, \tau) = \mathcal{N}\left(\mu \;\middle|\; \overline{z}, \frac{1}{N\tau}\right) \mathrm{Gamma}\left(\tau \;\middle|\; \frac{N-1}{2}, \frac{N}{2}(\overline{zz} - \overline{z}^2)\right).$$

This implies

$$\begin{aligned} \mathbb{E}_{q(\tau)}\left[\tau\right] &= \frac{(N-1)}{N(\overline{zz} - \overline{z}^2)}, \\ \mathbb{E}_{q(\mu,\tau)}\left[\mu\tau\right] &= \overline{z}\mathbb{E}_{q(\tau)}\left[\tau\right]. \end{aligned}$$

Finally, we derive the update equation for $q(\boldsymbol{z})$:

$$\begin{aligned} \ln q^*(\boldsymbol{z}) &= \mathbb{E}_{q(W,\mu,\tau)}\left[\ln p(X \mid \boldsymbol{z}, W) + \ln p(\boldsymbol{z} \mid \mu, \tau)\right] + \text{const} \\ &= \mathbb{E}_{q(W)}\left[\ln p(X \mid \boldsymbol{z}, W)\right] + \mathbb{E}_{q(\mu,\tau)}\left[\ln p(\boldsymbol{z} \mid \mu, \tau)\right] + \text{const}. \end{aligned}$$

We expand the terms $\ln p(X \mid \boldsymbol{z}, W)$ and $\ln p(\boldsymbol{z} \mid \mu, \tau)$. All expressions that are independent of $\boldsymbol{z}$ are treated as constants.

$$\begin{aligned} \ln p(X \mid \boldsymbol{z}, W) &= \sum_j \ln \mathcal{N}(\boldsymbol{x}_j \mid Z\boldsymbol{w}_j, \beta^{-1} I) \\ &= -\frac{\beta}{2} \sum_j \|Z\boldsymbol{w}_j - \boldsymbol{x}_j\|^2 + \text{const} \\ &= \beta \sum_i \left(-\frac{1}{2}\phi(z_i)^\top \sum_j \boldsymbol{w}_j \boldsymbol{w}_j^\top \phi(z_i) + \phi(z_i)^\top \sum_j X_{ij} \boldsymbol{w}_j \right) + \text{const}. \end{aligned}$$

$$\begin{aligned} \ln p(\boldsymbol{z} \mid \mu, \tau) &= -\frac{\tau}{2} \sum_i (z_i - \mu)^2 + \text{const} \\ &= \sum_i \left(-\frac{\tau}{2} z_i^2 + \tau\mu z_i \right) + \text{const}. \end{aligned}$$

Substituting these expansions into the variational update expression and taking the expectation yields:

$$\ln q^*(\boldsymbol{z}) = \sum_i \ln q^*(z_i),$$

where

$$\begin{aligned} \ln q^*(z_i) = {}& \beta \left(-\frac{1}{2}\phi(z_i)^\top \sum_j \mathbb{E}_{q(\boldsymbol{w}_j)}\left[\boldsymbol{w}_j \boldsymbol{w}_j^\top\right] \phi(z_i) + \phi(z_i)^\top \mathbb{E}_{q(W)}\left[W\right] \boldsymbol{x}_i^\top \right) \\ &+ \left(-\frac{1}{2}\mathbb{E}_{q(\tau)}\left[\tau\right] z_i^2 + \mathbb{E}_{q(\mu,\tau)}\left[\tau\mu\right] z_i \right) + \text{const}. \end{aligned}$$

Recall that $\boldsymbol{x}_i^\top$ is a column vector representing i-th row of the observation matrix X.

Because the update expression separates as a sum over i, the optimal factor $q(\boldsymbol{z})$ factorizes as $\prod_i q(z_i)$.

The update equations require expectations of the form

$$\mathbb{E}_{q(z_k)}\left[\phi_i(z_k)\phi_j(z_k)\right], \quad \mathbb{E}_{q(z_k)}\left[\phi_i(z_k)\right], \quad \mathbb{E}_{q(z_k)}\left[z_k\right], \quad \mathbb{E}_{q(z_k)}\left[z_k^2\right],$$

for all relevant indices i, j, k. Since the basis mapping is nonlinear, the optimal factor $q(z_i)$ is generally non-Gaussian and these expectations must be approximated numerically.